

ROAD CONDITION DETECTION USING AI-BASED COMPUTER VISION

Abstract:

Road infrastructure plays an important role in transportation safety, traffic management, and smart city development. Manual inspection of road conditions is time-consuming, expensive, and often inefficient for large-scale monitoring. The proposed project, **Road Condition Detection**, focuses on developing an AI-based computer vision system capable of automatically detecting, classifying, and segmenting different types of road damages and roadside elements from images or video streams.

The system aims to identify multiple road damage classes such as potholes, rutting, cracks, raveling/traveling defects, and other surface irregularities. In addition, the project also detects important road-related objects including trees, traffic signs, lane/markings lines, and security railings. Using deep learning models for object detection and semantic segmentation, the system can analyze road scenes in real time and provide accurate information about road conditions.

The project involves dataset collection and preprocessing, annotation of road defects, model training using convolutional neural networks and segmentation architectures, performance evaluation, and deployment for practical applications. Technologies such as Python, OpenCV, YOLO, and segmentation models like U-Net or Mask R-CNN can be utilized for implementation.

The developed solution can support government organizations, smart transportation systems, and road maintenance authorities by enabling faster damage assessment, reduced inspection costs, predictive maintenance, and improved public safety. This project contributes toward intelligent infrastructure monitoring and sustainable transportation management.

Project Objectives

1	Detect potholes, cracks, rutting, and other road damages automatically.
2	Classify different categories of road surface defects.
3	Perform segmentation of damaged regions for precise analysis.
4	Detect roadside objects such as traffic signs, trees, lane markings, and security railings.
5	Develop a scalable AI-based monitoring system for smart road maintenance.

Possible Technologies Used

- Python Programming
- OpenCV
- YOLO Object Detection
- U-Net / Mask R-CNN Segmentation Models
- TensorFlow / PyTorch
- Image Processing and Deep Learning Techniques